Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Chutjian, Ara Ph.D.		
Project Title:	Miniature Quadrupole Mass Spectrometer Array(QMSA)/Trace Gas Analyzer Flight Instrument		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ara.chutjian@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	818-354-7012
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	California Institute of Technology		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	90210	**Congressional District:**	28
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	12/01/2000	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS ; detection of ammonia and air leaks external to the ISS hull		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This work is directed at upgrading the performance of the QMSA and the Trace Gas Analyzer (TGA) for a reflight mission to the International Space Station. Upgrades include replacement of the CPU, ion pumps, addition of a getter pump, and addition of a new sunlight-visible display. This reflight is scheduled for October 2006. Work is being done jointly between JPL and Oceaneering Space Systems (Houston), with JPL as the PI.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The TGA can be used in Earth applications needing a field-deployable instrument. Such applications would be the monitoring of civilian and military facilities (air-sea cargo containers, airports, border crossings, military bases, nuclear power plants) against terrorist attack; as well as at superfunds cleanup sites, and for down-hole water table monitoring,
Task Progress:	Funds have been received in FY05 to select a new ion pump, getter pump, CPU, and sunlight-visible display. Funds for FY06, if granted, will be used to implement these changes, and to carry out QA&R tests on the TGA.
Bibliography Type:	Description: (Last Updated: 02/27/2004)